

Monday Warm Up: Capacitors, Current and DC circuits

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1 Memory Bank

- $Q = CV$... Relationship between capacitance, charge, and voltage.
- $C_{tot} = C_1 + C_2 + C_3 + \dots$... Capacitors in parallel.
- $\Delta U = QV/2 = (1/2)CV^2 = Q^2/(2C)$... Energy stored in a capacitor.
- $I = \Delta Q/\Delta t$... Definition of current.
- $\Delta V = IR$... Ohm's Law.

2 Capacitors

1. Consider Fig. 1 (top). *The sum of all the capacitances is equal to the total capacitance. Why?*
2. Consider Fig. 1 (bottom). *The sum of the reciprocals of the capacitances is equal to the reciprocal of the total capacitance. Why?*

3. Consider Fig. 1 (top). If C_1 , C_2 , and C_3 are each 50 nF, what is the total capacitance?

4. Consider Fig. 1 (bottom). If C_1 , C_2 , and C_3 are each 50 nF, what is the total capacitance?

5. What is the energy stored in Fig. 1 (top), if the individual capacitors are each 50 nF, and the voltage is 5V?

6. What is the energy stored in Fig. 1 (bottom), if the individual capacitors are each 50 nF, and the voltage is 5V?

3 Current and DC Circuits

1. What is the current in milliamperes produced by the solar cells of a pocket calculator through which 4.00 C of charge passes in 4.00 h?

2. A defibrillator sends a 6.00 A current through the chest of a patient by applying a 10 kV potential as in the figure below. What is the resistance of the path? The defibrillator paddles make contact with the patient through a conducting gel that greatly reduces the path resistance.

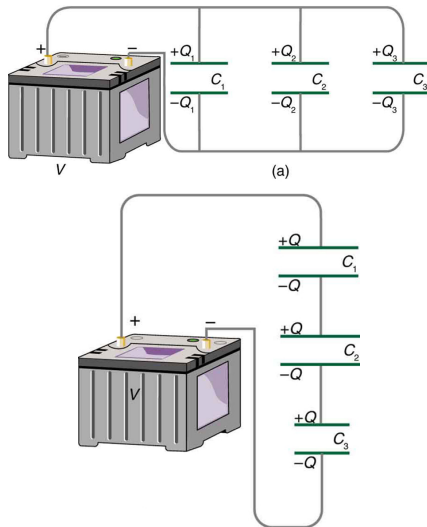


Figure 1: The relationship between potential energy and voltage.